Mid-West University

Examinations Management Office

End-Semester Examinations -2080

Bachelor level/ B.E. Civil/ 1st Semester

Time: 3 hours

Full Marks: 50

Pass Marks: 25

Subject: Applied Mechanics I (Statics) (CE411/CE101)

- *Attempt all the questions*
- *Figures in the margin indicate full marks.*
- *Assume suitable values, with a stipulation, if necessary.*
- *Candidates are required to answer the questions in their own words as far as possible.*

1. a) Explain transmissibility principle of Force. Determine the support reaction at contact point of given system. Assume contact surfaces are smooth. [1+4]

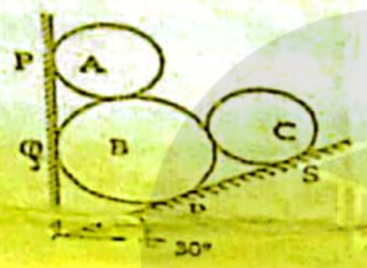
Take,

Weight of sphere A and C = 300N

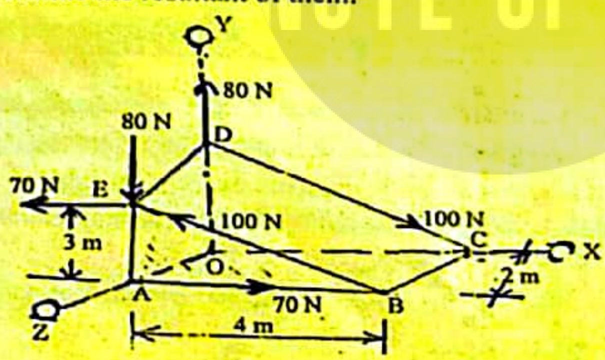
Weight of sphere B=600N

Diameter of A and C = 800mm

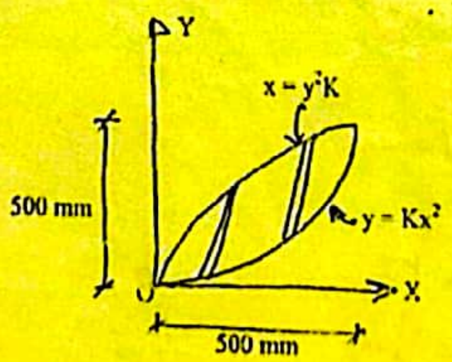
Diameter of B:1100mm



b) Three pairs of couples are acted on the triangular block as shown in figure below. Determine the resultant of them. [5]



2. a) Distinguish between center of gravity and centroid. Calculate Centroid of following geometry by direct integration method. [1+4]



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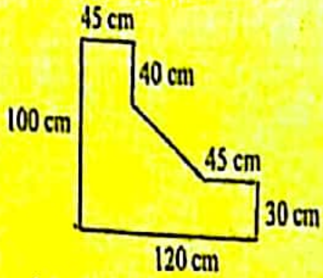
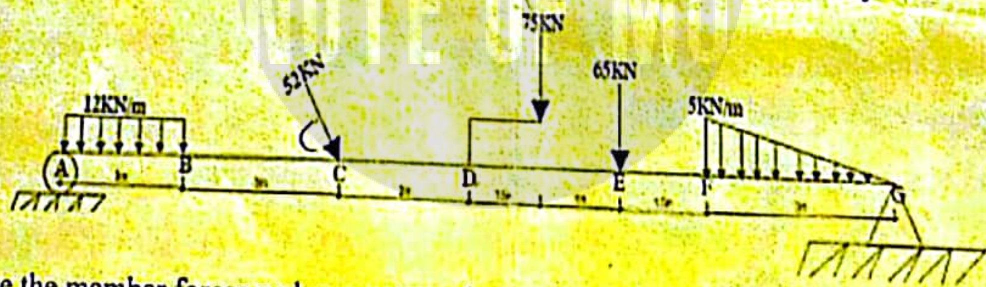
b) State the Parallel axis of theorem. Calculate the moment of inertia of following regular geometry about centroidal axis.



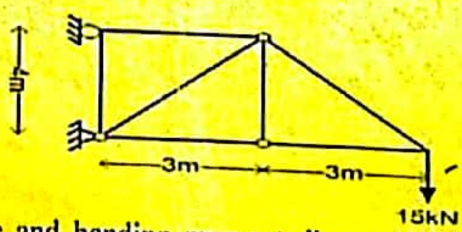
3. a) The position of the particle is defined by the relation $X=3t^3-7t^2+12$ where t expressed second and X expressed in cm. Determine: [5]
i. The time when velocity is zero
ii. The position and acceleration at that time
iii. The total distance travelled by the particle.

b) List out law of frictions. Two masses m_1=25kg and m_2=15kg are tied together by a rope parallel to the inclined plane surface as shown in figure. If μs for m_1 is 0.25 and that for m_2 is 0.5 find (a) the value of θ for which masses will just start slipping. (b) tension in the rope. [5]

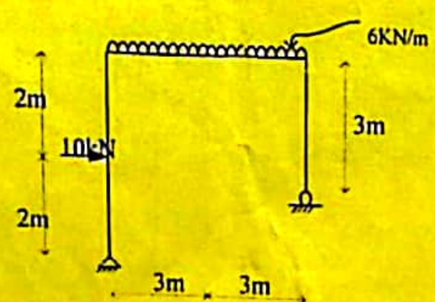
4. Derive the relation between load intensity, share force and bending moment. And draw the share force axial and bending moment diagram of following beam. 52KN load is 60-degree inclination and dimension AB=3m, BC=3m, CD=2m, DE=2.5m, EF1.5m and GF=3m respectively. [3+7]



5. a) Calculate the member forces and support reaction of following truss by joint method. [5]



b) Draw the Axial force, shear force and bending moment diagram of the following frame given below [5]



THE END

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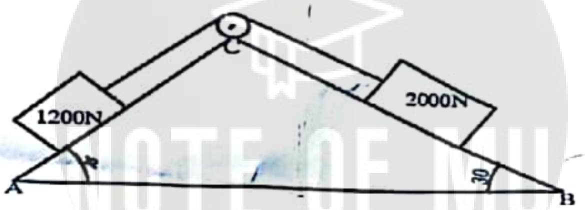
MID WEST UNIVERSITY

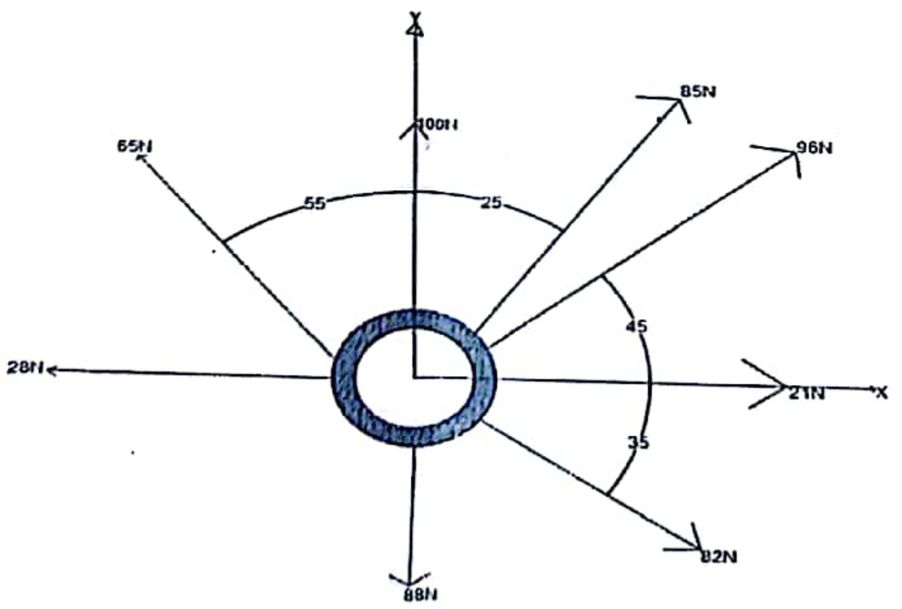
Level: Bachelor	Mid Examination	Year : 2023
Programme: BE Civil		Full Marks: 25
Course: Applied Mechanics-I (Statics)		Pass Marks: 12.5
		Time : 1.5Hr

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

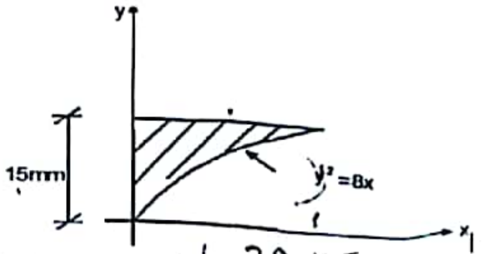
1 Two planes AC and BC are inclined at [illegible]° and 30° to the horizontal. A load of 2000N rests an inclined in plane BC and 1200N on plane AC as shown in figure below. Determine the Value of Ø for the equilibrium. Take coefficient of friction (μ) for plane AC as 0.25 and that of BC as 0.35. 5



2 State the Principle of Transmissibility. Calculate the resultant and direction of the following system of forces acts on a ring also draw resultant arrow diagram 5



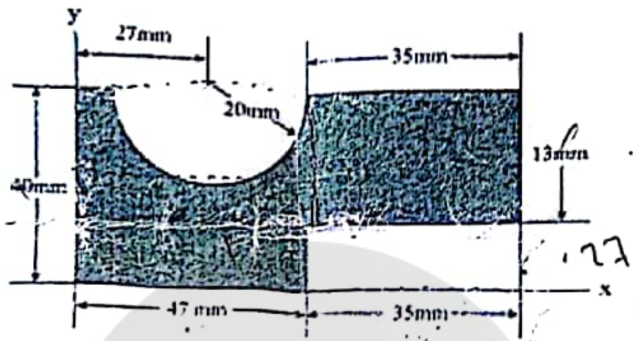
3 State Parallel Axis Theorem and Perpendicular Axis Theorem and determine the MOI about centroidal Axis of the following geometry by direct integration method. 1+ 1+ 4



$\frac{152}{8} = ?$

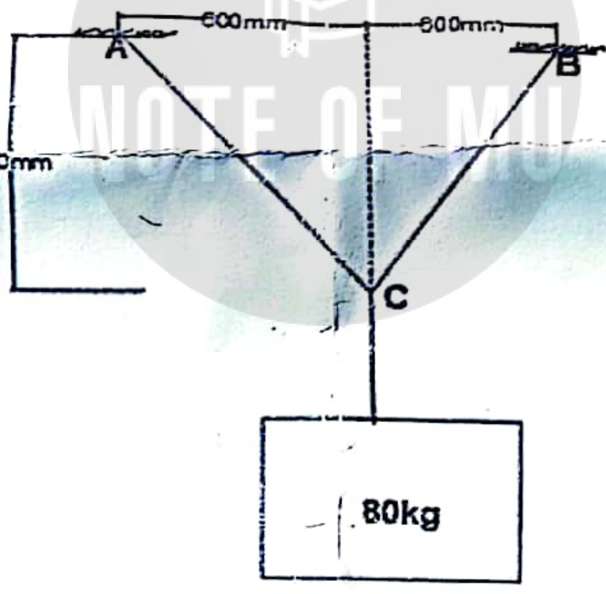
28.125

4 Distinguish between center of gravity and centroid. Locate the Centroid of given regular Geometry. 2+ 4



.27

5 Calculate The Tension Forces on links. 3



NOTE OF MU

$x = \frac{y^2}{8}$

$I_{xx} = I_{cm} + (x_1 - \bar{x})A$

D

Mid-West University

Examinations Management Office

Surkhet, Nepal

Final Examinations – 2079

elor level/ Civil engineering/ First Semester
e: 3 hrs
ject: Applied Mechanics I (CE 411/CE101)(Statics)

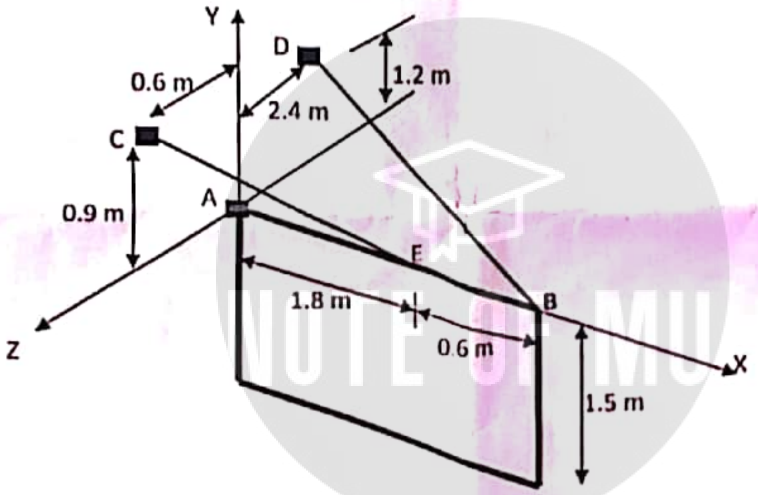
Full Marks: 50
Pass Marks: 25

- *Candidates are required to answer the questions in their own words as far as possible.*
- *Assume suitable values, with stipulation, if necessary.*
- *Figures in margin indicate full marks.*

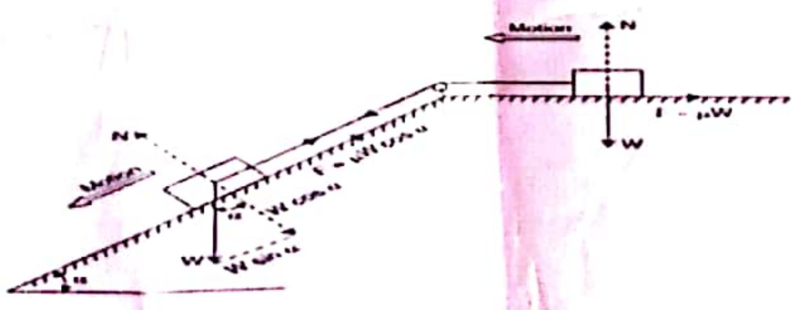
Attempt all the questions

State Varignon's Theorem also defines Kinetics and Kinematics with its significance in engineering.

A 120 Kg sign of uniform density measures 1.5 by 2.4 m and is supported by a ball and socket at A and by two cables. Determine the tension in each cable



Two blocks of equal weights and connected by a string passing over a frictionless pulley, rest on surfaces; one on a horizontal surface and the other on an inclined surface as shown in Fig. For both the surfaces the co-efficient of static friction $\mu = 0.2$. For what value of angle α the motion of the two blocks will impend?



What is statically determinate and indeterminate structure, write with example. What is the Physical meaning of equilibrium and its essence in structural application?

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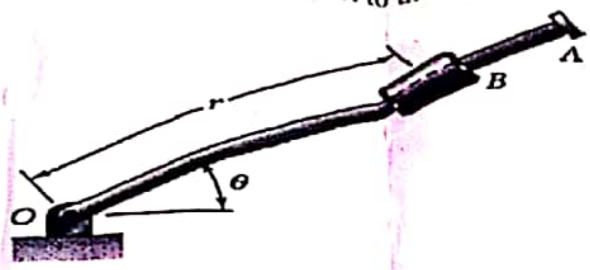
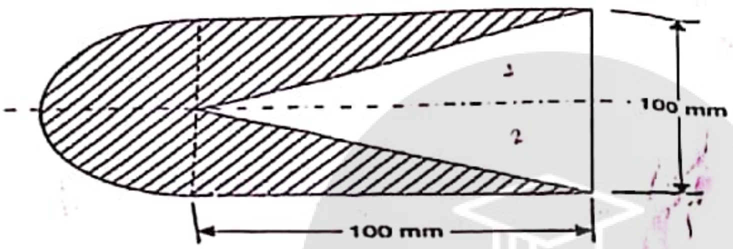
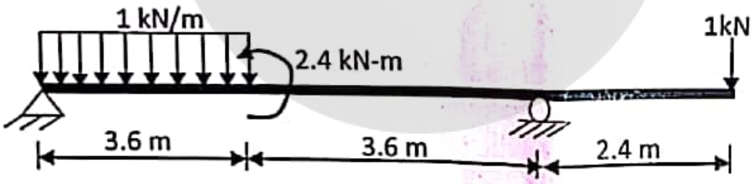
5. Rotation of the arm about O is defined by $\theta = 0.15t^2$ where θ is in radians and t in seconds. C B slides along the arm such that $r = 0.9 - 0.12t^2$ where r is in meters. After the arm has rotate through 30°, determine
(a) the total velocity of the collar,
(b) the total acceleration of the collar, and
(c) the relative acceleration of the collar with respect to the arm.



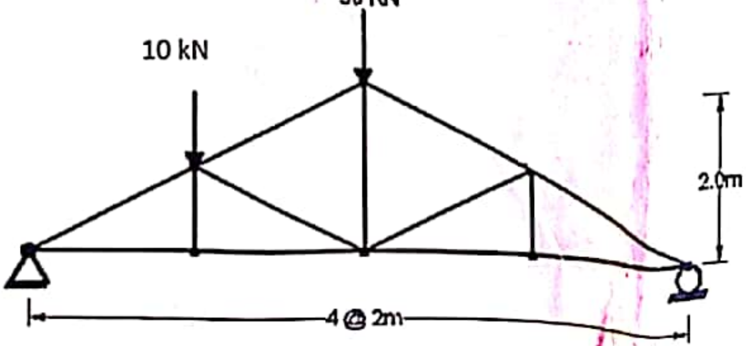
6. Determine the Centre of gravity of a homogeneous solid body as shown in Fig.



7. Draw shear force and bending moment diagram of the following beam



8. Determine the force in each member of following truss by using joint method.



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Mid-West University

Examinations Management Office

End semester Examinations -2078

Bachelor level/ B.E. Civil/1st Semester | Full Marks : 50

Time: 3 hrs | Pass Marks : 25

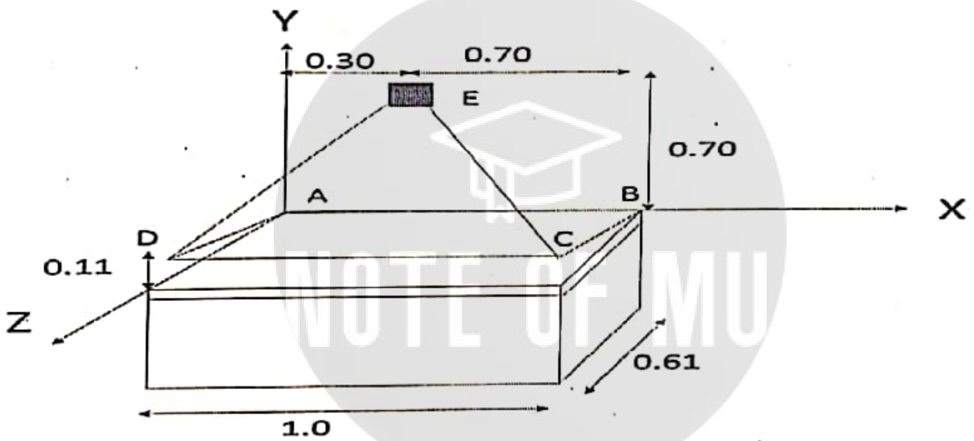
Subject : Applied Mechanics(CE411/101)

- *Candidates are required to answer the questions in their own words as far as possible.*
- *Assume suitable values, with stipulation, if necessary.*
- *Figures in margin indicate full marks.*

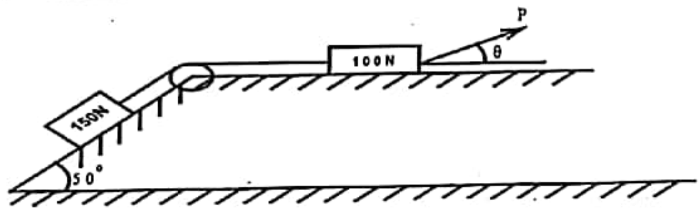
Attempt all the questions

1) Define Statics, Dynamics, Kinetics and Kinematics with its significance in engineering. 4

2) The 0.61x1 m lid ABCD of storage bin is hinged along side AB and is held open by looping cord DEC over a friction less hook at E. if the tension in the cord is 66N determine the moment about each of the coordinate axis of the force exerted by the cord at D. 6



3) Referring the figure shown, determine the least value of the force P to cause motion to impend rightwards. Assume the coefficients of friction under the blocks to be 0.2 and consider pulley as frictionless. 6



4) What type of structure is statically determinate and indeterminate structure, write with example. What is the Physical meaning of equilibrium and its essence in structural application? 3+1

5) The rotation of the 0.9 m arms OA about O is defined by the relation $\theta = 0.18t^2$ where θ is expressed in radian and t in second. Collar B slides along the arm in such a way that its distance from O is $r = 0.8 - 0.12\ t^2$, r in meter and t in second. After the arm OA rotated through 30° determine 4
   a) Total velocity of the collar
   b) Total acceleration of the collar

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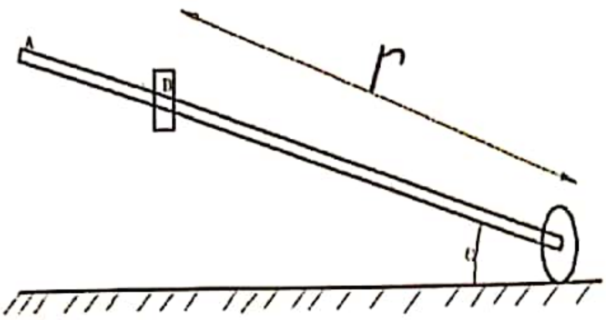
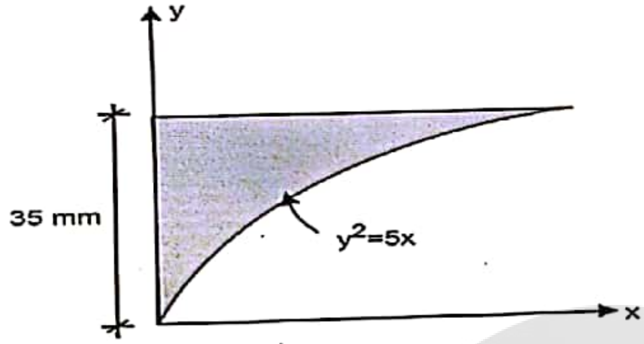
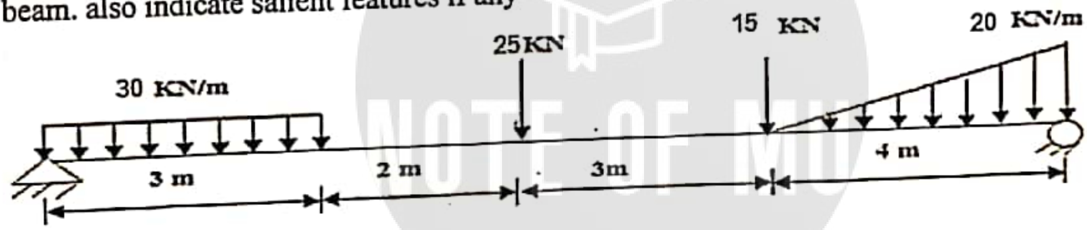


6) Define the Statements of parallel and perpendicular axis theorem By direct integration method, find the co-ordinates of centroid of the shaded area with respect to given axes. 2+6

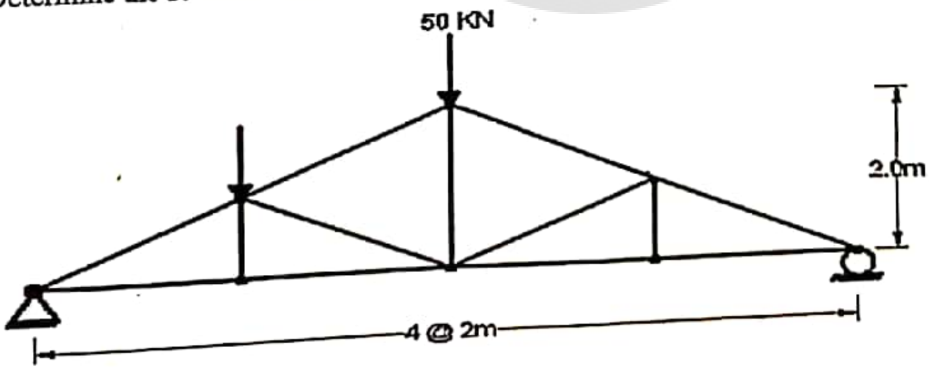


7) Derive the relationship between load intensity, shear Force and bending moment with their usual meanings. Draw axial force, shear force and bending moment diagram of given simply supported beam. also indicate salient features if any 3+7



NOTE OF MU

8) Determine the force in each member of following truss by using section joint method. 8



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MIDWESTERN UNIVERSITY
FACULTY OF ENGINEERING
Examination Control Division

Exam	Regular/Back		
Level	BE	Full Marks	50
Program	Civil/Hydro	Pass Marks	25
Year/Part	I/I	Time	3 Hrs.

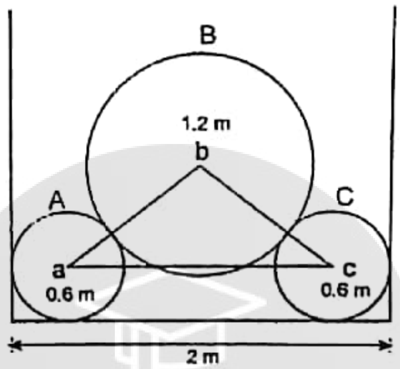
2074 Chaitra

Subject: Applied Mechanics I (Statics) (CE101)

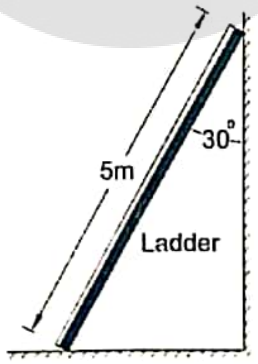
- *Candidates are required to answer the questions in their own words as far as possible.*
- *Assume suitable values, with stipulation, if necessary.*
- *Figures in margin indicate full marks.*

1 (a) Describe briefly the concept of particle, rigid body and deformed body. Define resultant couple. [3+2]

(b) The cylinders A and C weigh 1000 N each and the weight of cylinder B is 2000 N. Determine the forces exerted at the contact points. (Diameter of B=1.2m & Diameter of A & C = 0.6m) [5]



2 (a) A uniform ladder of weight 250 N and length 5m is placed against a vertical wall in a position where its inclination to the vertical is 30^0. A man weighting 600N climbs the ladder. At what position will be induce slipping? Take coefficient of friction u=0.25 at both the contact surface of the ladder. [5]



(b) Explain the principle of transmissibility with neat sketch. Write short notes on couple and moment. [2+2=4]

3 (a) Determine the moment of inertia of a the given I section shown in figure below about the centroidal axes x-x and y-y. [6]

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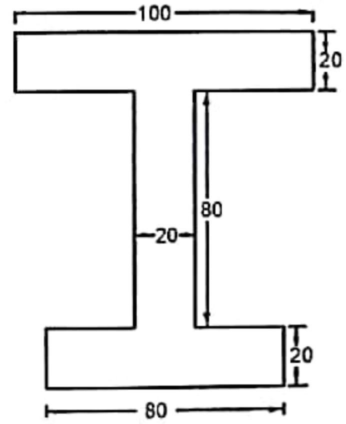
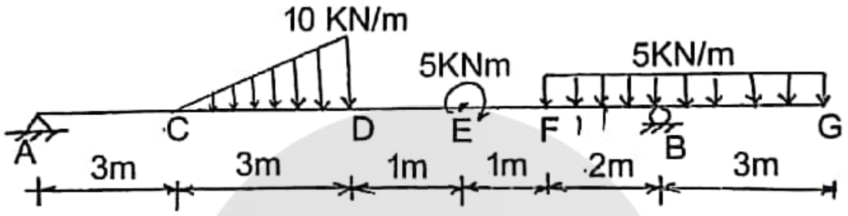


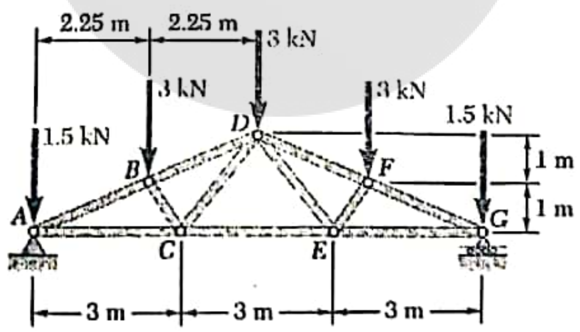
(b) State and prove the parallel axis theorem with neat sketch. [5]

4 (a) Draw the shear force and bending moment diagram for the given beam loaded as shown in figure. Indicate maximum bending moment. [6]



(b) Derive the relationships among load, shear force and bending moment with neat sketch. [4]

5 (a) Determine the force in each member of the given roof truss shown. State whether each member is in tension or compression. [5]



(b) The motion of a particle is defined by the relation $x = t^2 - 10t + 30$ where x is expressed as meter and t as second. Determine: [5]

a) the time when velocity becomes zero

b) velocity of particle at t=5sec

c)position and total distance travelled by particle when t= 8 sec.

NOTE OF MU

Mid-Western University

Examinations Management Office

Chance Examinations -2070

Full Marks 50
Pass Marks 25

Bachelor level: B.E. Civil I Semester
Time: 3 hrs
Subject : Applied Mechanics-I statics (CE 1?)

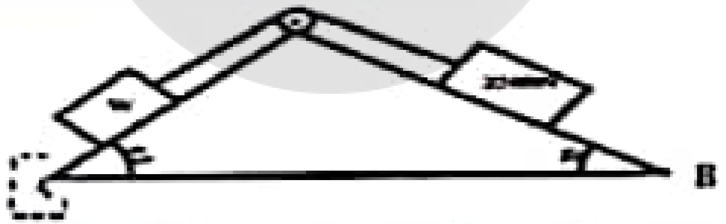
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

1. a. Two Cylinders A and B rest in a channel as shown. A has diameter of 220m and weight 25 Kg B has diameter of 150mm and weight 20 Kg. The channel is 250mm wide at the bottom with one side vertical and other side at 120° as shown. Determine the forces and reactions at all four points of contact. 5

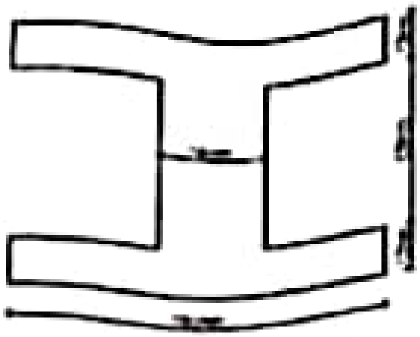
NOTE OF MU

b. Two planes AC and BC are inclined at 60° and 30° to the horizontal. A load of 2500N rests an inclined in plane BC and W on plane AC as shown in figure below. Determine the range of W for the equilibrium. Take coefficient of friction (μ) for plane AC as 0.70 and that of BC as 0.35. 5



2. a. Determine the moment of inertia of the given section about x-axis and y-axis by direct integration method. 5

b. State Parallel Axis Theorem and Perpendicular Axis Theorem and determine the centroid of the following geometry. 2+3



3

a. Explain transmissibility principles and verify dynamic equilibrium equation with suitable example 5

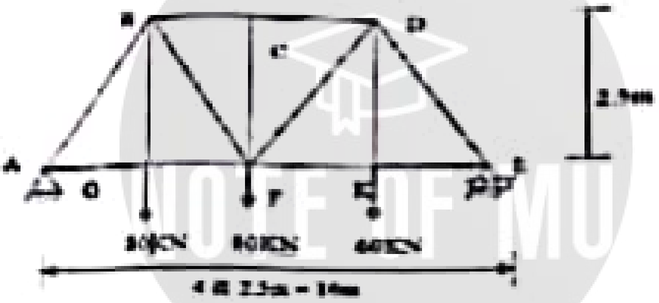
b. The motion of a particle is defined by the relation $x=6t^4-12t^3+25$ where x is expressed in meters and t in seconds. Determine the time, position and acceleration of a particle when v=0 and 2m/s (v is velocity of a particle) 5

4

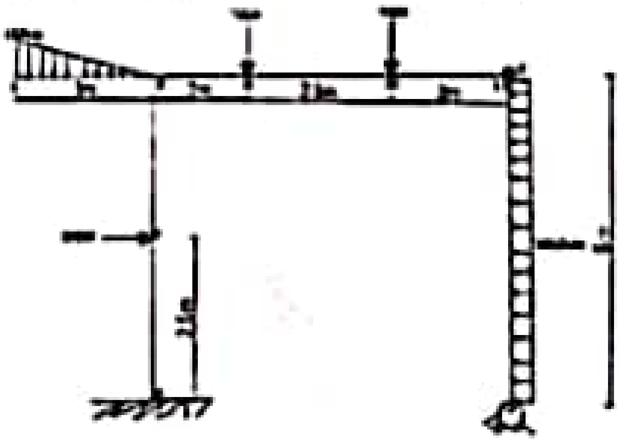
a. Derive the relationship between load intensity, shear force and bending moment with their usual meaning. Draw the axial force, shear force and bending moment diagram of the following beam. 1+5

5

a. Calculate the member forces of the following truss by joint method 5



b. Calculate support reaction and Draw BMD and SFD and axial force diagram of following truss 5



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MID-WESTERN UNIVERSITY
FACULTY OF ENGINEERING
CENTRAL CAMPUS OF ENGINEERING

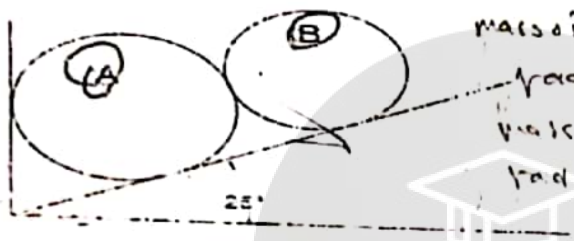
DEPARTMENT OF HYDROPOWER ENGINEERING

Level: Bachelor Assessment	Year : 2020
Programme: Hydropower	Full Marks: 25
Course: Applied Mechanics-I(statics)	Pass Marks: 12.5
	Time 1.5Hr.

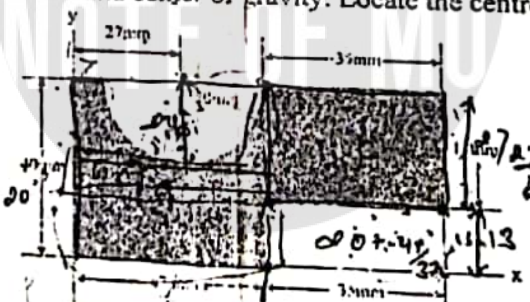
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

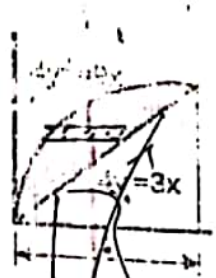
1 State principle of transmissibility. Two spheres A and B are placed in a container as shown in figure. Determine the reaction from the walls of the container. Mass of sphere A is 20kg and radius is 15cm. Mass of sphere B is 16kg and radius is 12cm. Assume all contact surfaces are smooth surfaces. 1+4



2 List out difference between centroid and center of gravity. Locate the centroid in the given figure. 1+4

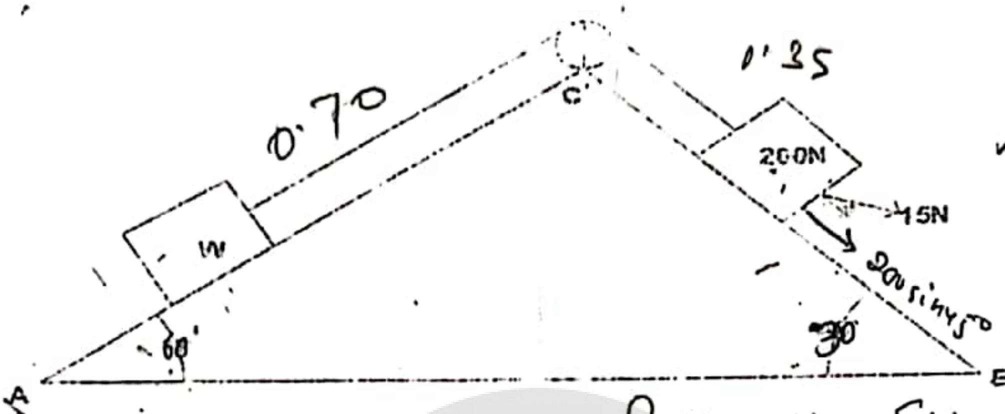


3 By direct integration method, find moment of inertia of the shown shaded area with respect to given axes. 5



$w + w \sin 60$

4 Two planes AC and BC are inclined at 60° and 30° to the horizontal. A load of 200N rests an inclined in plane BC and W on plane AC as shown in figure below. Determine the range of W for the equilibrium. Take coefficient of friction (μ) for plane AC as 0.70 and that of BC as 0.35. 5



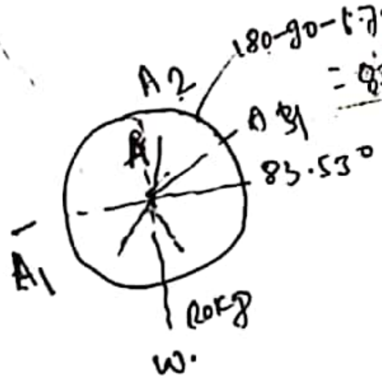
$\mu_2 = 0.35$

$\mu_1 = 0.70$

$w, 200N$

$R = w_1 \sin 60$

5 A 1.5m×2.4m sign board of uniform density weighs 1500N and is supported by the ball and socket joint at A and by two cables. Determine the tension in each cable and the reaction at A. 5



Best of Luck!!

THE END!

$\Sigma f_y = 0$ ↑+ve

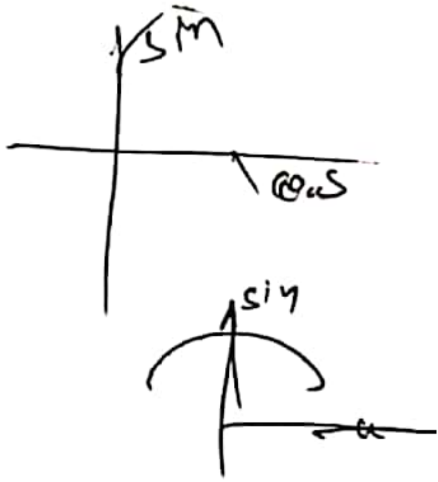
$A_2 \cos 80 - A_3 \sin 83.53 - W = 0$

$A_2 \cos 80 - \sin 0.895 = 0$

$A_2 \cos 20 = 0.89$

$A_2 = \cos(2)$

$A_2 = 0.94$



NOTE OF MU

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